

LIFE PROCESSES

Student Learning Pack

1. Metadata

Topic	Life Processes
Level	Class X
Lesson Length	60 minutes

Scope Note

Focus on the definition, importance, and main components of life processes: nutrition, respiration, excretion and criteria for life.

2. Learning Objectives

- LO1: Explain what life processes are and why they are essential for maintaining organismal order.
- LO2: Describe how organisms obtain energy and raw materials through nutrition.
- LO3: Explain how nutrition provides the energy needed for continual molecular movement.
- LO4: Note that excretory processes are not covered in the provided excerpt.
- LO5: Identify the criteria for life discussed in the excerpt, focusing on molecular movement and maintenance of order.
- LO6: Summarise how nutrition supports life processes as described in the excerpt.

3. Prerequisite Knowledge

- Basic cell structure and the concept that cells are made of molecules.
- Understanding of diffusion as a passive transport mechanism.
- Awareness of the fundamental needs of living organisms (food, water, air).

4. Key Terms

Term	Definition
Life processes	The set of maintenance functions (e.g., nutrition, respiration, excretion) that keep an organism alive and ordered.
Nutrition	The intake and processing of food to obtain energy and raw materials needed for growth and maintenance.
Diffusion	Not discussed in the provided textbook excerpt.
Respiration	Not discussed in the provided textbook excerpt.
Excretion	Not discussed in the provided textbook excerpt.
Molecular movement	Invisible movements of molecules inside cells that are essential for maintaining order.

Term	Definition
Raw materials	Not discussed in the provided textbook excerpt.
Transport system	Network (e.g., circulatory system) that moves nutrients, gases and wastes between cells and specialised tissues.

5. Key Points

- Life processes are the maintenance activities that keep an organism's ordered structure alive.
- Invisible molecular movement is essential for life because it enables continual repair and order.
- Nutrition supplies the external energy and carbon-based raw material needed for life processes.
- Most food sources are carbon-based, matching the chemistry of living organisms.
- Beyond nutrition, the excerpt does not address additional life processes.

6. Real-World Applications

Analyzing a Food Label

Students examine a packaged food label to identify carbon-based nutrients that provide energy and raw material.

Think about: Which components on the label are carbon-based sources of energy, and how do they support the organism's life processes?

Suggested time: 5 minutes

Observing Plant Growth with/without Fertiliser

Students grow two identical plants, one receiving a carbon-based fertiliser and one without, to see the effect on growth and maintenance.

Think about: How does the added carbon-based material influence the plant's ability to carry out life processes?

Suggested time: 7 minutes

7. Core Notes

What are Life Processes?

Living organisms are organised from tissues down to molecules. Over time the environment tends to break down this order. To stay alive, organisms must constantly repair and maintain their structures. The set of activities that achieve this continual repair is called life processes. They are required even when an organism appears inactive, such as during sleep.

Molecular Movement and Life

Molecules inside cells are never static; they move, collide, and are transported. This molecular motion allows the rebuilding of damaged structures and the transport of nutrients and waste. Viruses lack observable molecular movement until they infect a cell, which is why their status as "alive" is debated.

Nutrition: Getting Energy and Raw Materials

Life processes require energy to prevent damage and breakdown. That energy comes from outside the organism as food. Nutrition is the process that transfers this external energy into the body. Because most life on Earth is carbon-based, the food we consume is also made of carbon-based molecules. The complexity of these carbon sources determines how different organisms process them.

Carbon-Based Food Sources

The molecules that make up living structures contain carbon. Consequently, the raw materials taken in as food are also carbon-based. Organisms break down these carbon molecules to obtain both energy and the building blocks needed for growth and repair.

Other Essential Life Processes

The excerpt focuses on nutrition as the primary life process; respiration and excretion are not discussed in this section.

8. Activities

Concept-Map of Life Processes (12 min)

Students create a concept map linking life processes (nutrition, respiration, excretion) with molecular movement and organismal order.

Debate: Are Viruses Alive? (10 min)

In small groups, students discuss the textbook claim that lack of molecular movement makes viruses controversial as living entities, then present arguments.

9. Questions

Q1. What are life processes and why are they necessary for all living organisms?

Q2. Why is nutrition essential for maintaining order in living organisms?

Q3. What raw materials do organisms obtain from outside and how are they used?

Q4. How does the excerpt describe the role of nutrition in maintaining organismal order?

Q5. According to the excerpt, what criteria are used to determine if something is alive?